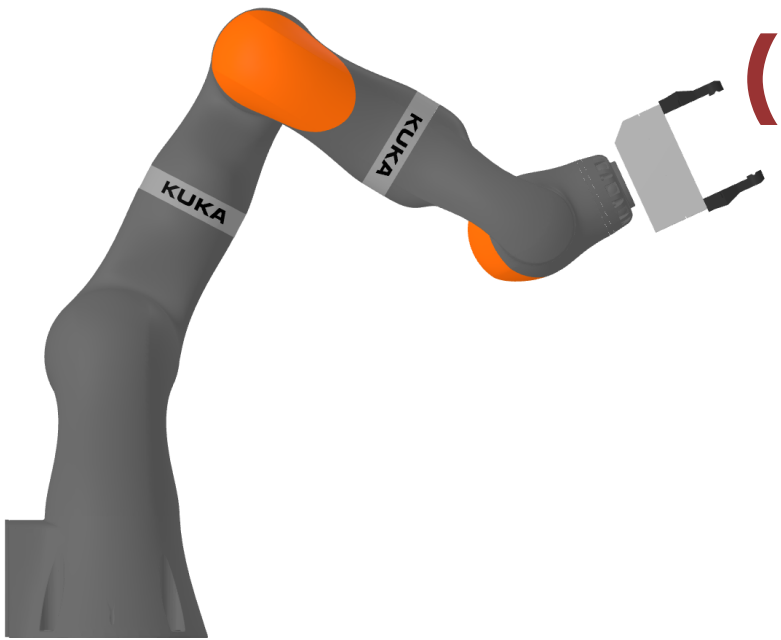
Basic Pick and Place (part 3)



MIT 6.4210/2: Robotic Manipulation

Fall 2025, Lecture 5

Massachusetts Institute of Technology

Kinematics

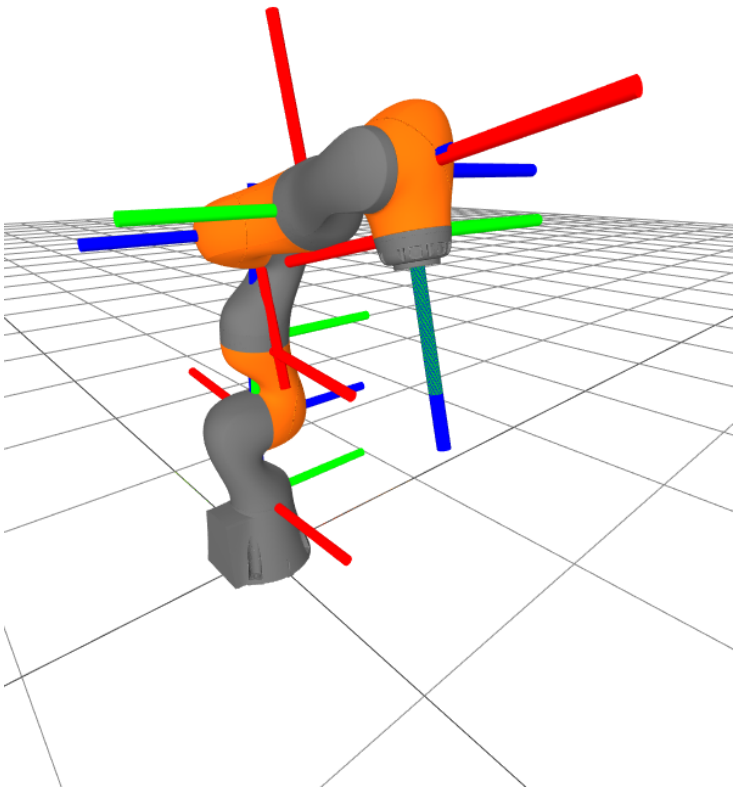
- *Forward kinematics*: joint positions ⇒ pose $q \Rightarrow X^B$
- *Inverse kinematics**: pose ⇒ joint positions $X^B \Rightarrow q$

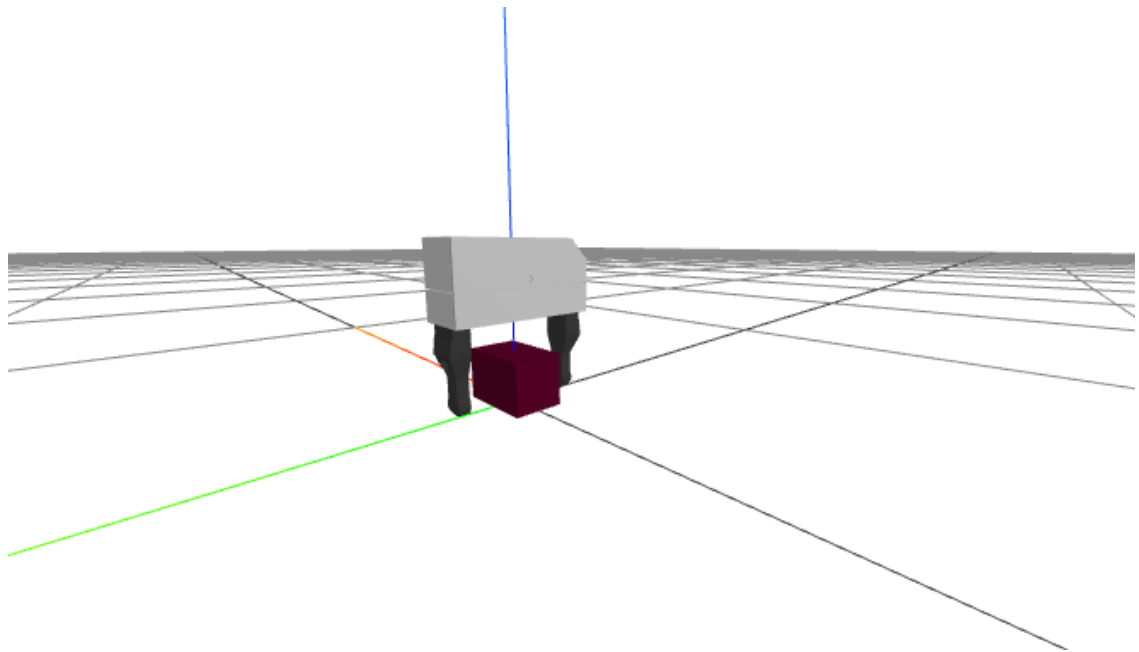
- *Differential kinematics*:
  joint positions, velocities ⇒ spatial velocity $q, v \Rightarrow V^B$

- *Differential **inverse** kinematics*:
  spatial velocity, joint positions ⇒ joint velocities
  $V^B, q \Rightarrow v$

Basic Pick and Place

Step 1: Kinematic Frames / Spatial Algebra

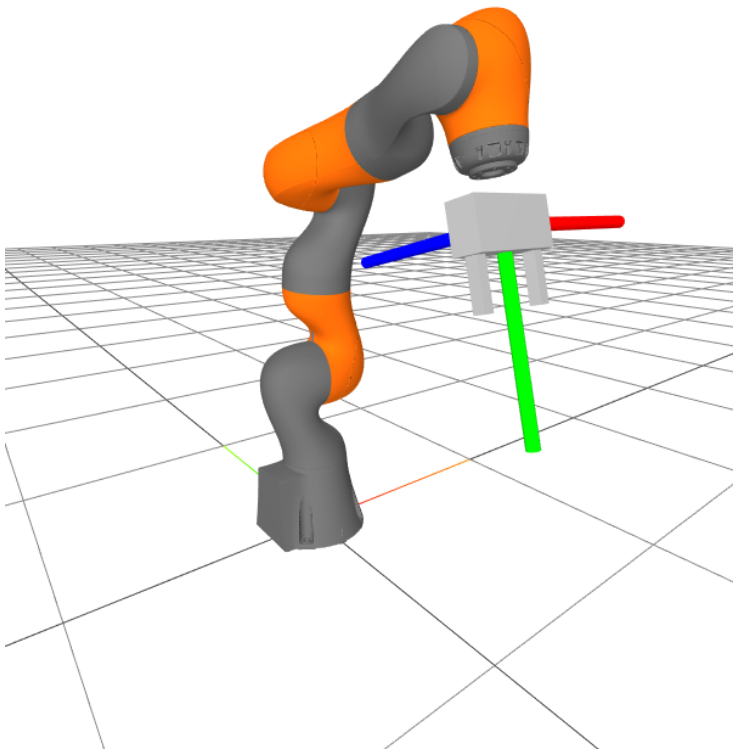


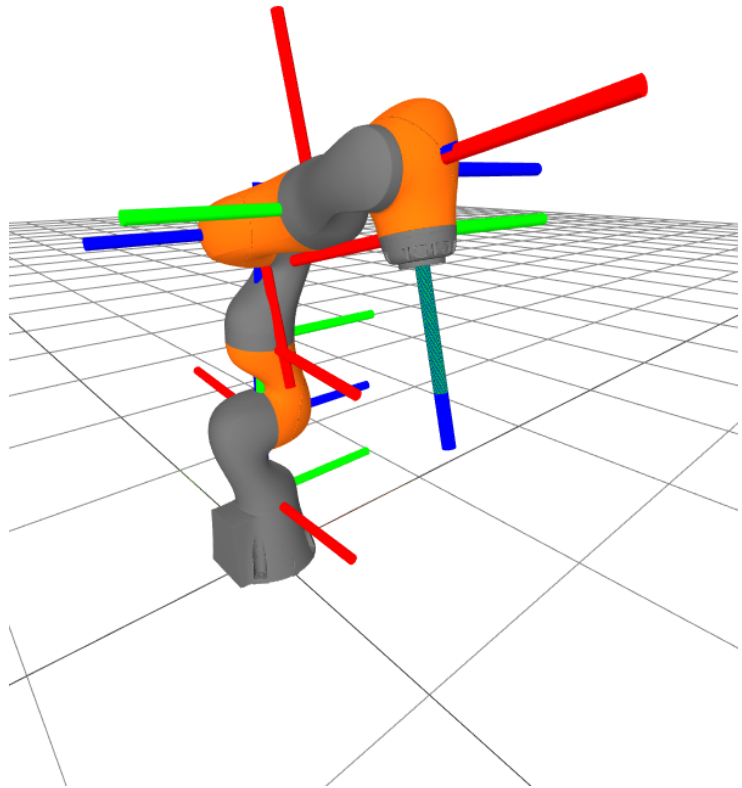


Step 2: Gripper Frame Plan "Sketch"

https://manipulation.mit.edu/data/pick_sketch.html

Step 3: Forward kinematics of the iiwa + WSG





Step 4: Differential Inverse Kinematics

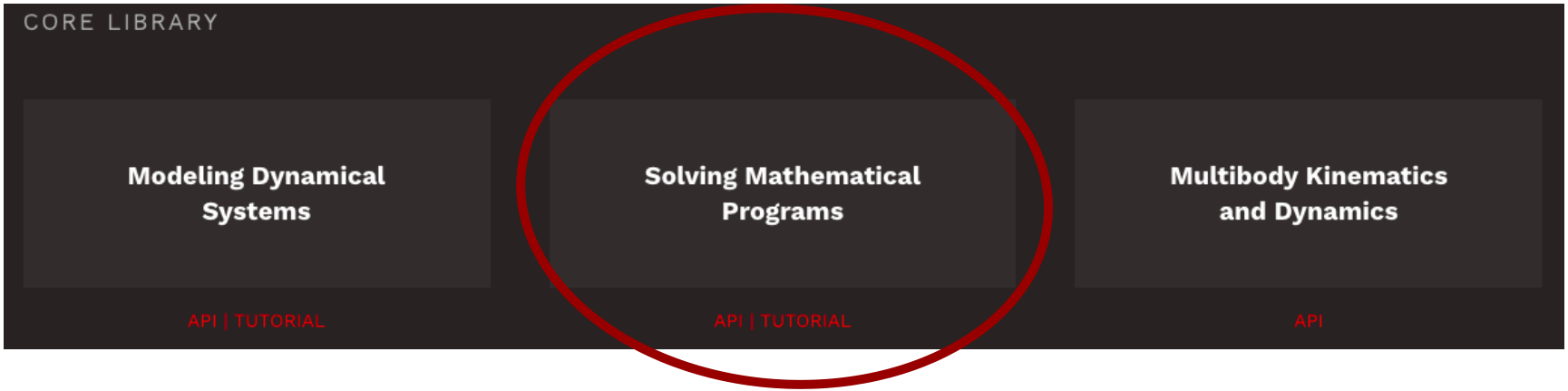
Kinematic Jacobian: $^{W}V^{G} = J^{G}(q)v$

spatial velocity

generalized velocity



$$v = [J^{G}(q)]^{+W}V^{G}$$




```
1 prog = MathematicalProgram()
2 x = prog.NewContinuousVariables(2)
3 prog.AddConstraint(x[0] + x[1] == 1)
4 prog.AddConstraint(x[0] <= x[1])
5 prog.AddCost(x[0] ** 2 + x[1] ** 2)
6 result = Solve(prog)
```

https://manipulation.mit.edu/pick.html#diff_ik_w_constraints